

Differentially Private Preference Data Synthesis for Large Language Model Alignment

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Abstract

Preference alignment is a crucial post-training step for large language models (LLMs) to ensure their outputs align with human values, but post-training on real human preference data raises privacy concerns, as these datasets often contain sensitive user prompts and human judgments. This paper makes three contributions toward closing this gap. **First**, we propose **DPPrefSyn**, and—to the best of our knowledge—the first algorithm to generate differentially private (DP) synthetic preference data for LLM alignment; DP-PrefSyn is a principled framework grounded in the Bradley–Terry preference model and the intrinsic geometric structure of pairwise human preference data, which learns an underlying preference model from private data under formal DP guarantees and then reuses the learned model together with public prompts to synthesize high-quality preference data. **Second**, we contribute two design ideas that make private preference modeling accurate under tight budgets: exploiting the shared linear structure of per-cluster reward models to capture heterogeneous human preferences, and applying DP Principal Component Analysis (DP-PCA) to improve learning accuracy. **Third**, through extensive experiments we demonstrate that DPPrefSyn attains competitive—and often superior—alignment performance under strong DP guarantees. Together, these contributions establish DP synthetic preference data as a practical, reusable alternative for privacy-preserving preference alignment across a broad range of applications. Our code will be released upon acceptance.

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1. Introduction

Preference alignment techniques, such as RLHF (Stiennon et al., 2020) and DPO (Rafailov et al., 2024), are now standard tools for aligning LLMs with human expectations. These algorithms rely on pairwise preference data, where human annotators compare two responses and select the one that better addresses the given prompt, and this data is then used to fine-tune LLMs by encouraging the model to rank preferred responses over less preferred ones (Ziegler et al., 2019; Ouyang et al., 2022; Bai et al., 2022). The effectiveness and reliability of preference alignment have driven its adoption in high-impact applications such as chat assistants (Achiam et al., 2023), mathematical reasoning tools (Shao et al., 2024), and code generators (Shen et al., 2023)—settings in which the underlying preference data is frequently sensitive.

Aligning LLMs with human preference data, however, raises substantial privacy concerns, since these datasets frequently include real user prompts and human feedback. Prompts may disclose personal information related to health, identity, or other sensitive topics, and feedback may reveal private beliefs, preferences, or behavioral patterns (Li et al., 2023; Yu et al., 2024). Prior works (Chowdhury et al., 2024; Yu et al., 2024; Wu et al., 2023a) mitigate these risks using the rigorous safeguards of differential privacy (DP) (Dwork et al., 2006b) and report encouraging results, yet they leave three concrete gaps that motivate our contributions. (i) **Partial protection**: many approaches privatize only part of the data—either the user prompts (Yu et al., 2024) or the preference labels (Chowdhury et al., 2024; Zhang et al., 2025), but not both—leaving room for adversaries to infer sensitive attributes from the unprotected component. (ii) **Method-specific design**: such works usually target a single post-training algorithm, such as RLHF (Wu et al., 2023a), and are not compatible with newer methods like DPO. (iii) **Data scarcity**: these methods are constrained by limited private preference datasets and often fall short of high-quality alignment, because collecting human preference annotations is costly and hard to scale. Together, these three gaps lead to the central question of this work:

Is it possible to achieve superior preference alignment while ensuring thorough privacy protection and maintaining

compatibility with emerging alignment methodologies?

We provide an affirmative answer. Our answer takes the form of four contributions, each of which directly targets one of the gaps above:

- **Contribution 1 (A new algorithm).** We introduce DP-PrefSyn (Figure 1), the first algorithm to generate DP synthetic preference data for LLM alignment. Unlike prior privacy-preserving alignment methods that protect only prompts or only labels, DPPrefSyn synthesizes an entirely new preference dataset and therefore protects prompts, responses, and preference labels at once. It is a principled framework grounded in the Bradley–Terry preference model and the intrinsic geometric structure of pairwise human preference data, and it learns an underlying preference model from private data under DP before reusing that model, together with public prompts, to synthesize high-quality preference data.
- **Contribution 2 (Three design ideas for accurate private modeling under tight budgets).** We contribute three complementary design elements that make DPPrefSyn accurate under strong privacy constraints. (a) *Capturing diverse human preferences via a shared linear reward structure and clustering:* the Bradley–Terry model with a linear reward lets us group preference samples by the embedding difference between concatenated prompt–preferred and prompt–dispreferred pairs, and approximate preferences within each cluster using a shared reward function. (b) *Improving learning efficiency through DP-PCA:* to counter sample inefficiency and clustering instability in high-dimensional spaces, we project the embeddings into a lower-dimensional space under DP while preserving key preference signals. (c) *Saving privacy budget with public prompts:* we generate candidate responses from public prompts and use private reward models only to select preference pairs, concentrating the entire privacy budget on modeling private preferences.
- **Contribution 3 (A rigorous privacy guarantee).** We prove that DPPrefSyn satisfies (ϵ, δ) -DP. Our analysis allocates a budget ϵ_0 to DP-PCA (Amin et al., 2019) for dimensionality reduction and ϵ_1 to DP-KMeans (Su et al., 2016) for clustering, and spends the remaining budget training per-cluster reward models with DP-SGD, whose cost we compose tightly with the Privacy Random Variable (PRV) accountant (Gopi et al., 2021). A key consequence of this guarantee is reusability: by the post-processing property of DP, the resulting synthetic dataset can be reused across multiple preference alignment methods and different LLMs at no additional privacy cost.
- **Contribution 4 (Strong empirical evidence).** We evaluate DPPrefSyn on standard benchmarks—question answering from OpenAssistant (Köpf et al., 2023) and

Anthropic-HH (Bai et al., 2022), and TL;DR summarization (Stiennon et al., 2020)—and show it delivers strong privacy while achieving competitive, and frequently superior, utility. For instance, with $\epsilon = 2$, DPO fine-tuning of Pythia-2.8B on our synthetic data reaches a GPT-4o win rate of 56.48% on Anthropic-HH, far above the 37.02% obtained by fine-tuning on real data. Crucially, and unlike method-specific baselines, DPPrefSyn plugs into various preference optimization methods (e.g., DPO and RLHF) and remains effective under strong privacy constraints.

2. Related Work

Differentially Private Alignment of LLMs. We position our first contribution against prior privacy-preserving alignment work, which each address only a subset of the gaps we target. Wu et al. (2023a) introduce a DP framework that aligns LLMs with reinforcement learning by adapting PPO to the DP setting; however, their approach is *limited to PPO*, whereas preference alignment continues to evolve with new methods. Chowdhury et al. (2024) study reward estimation from preference-based feedback using label-DP (Ghazi et al., 2021) to protect human annotators, and Zhang et al. (2025) propose AUP-RLHF, a user-level label-DP framework for RLHF. While both safeguard preference labels effectively, they *do not address the sensitivity of prompts or responses themselves*, and both are *constrained by limited private preference data*, so they may not reach high-quality alignment given how costly human annotations are. Yu et al. (2024) instead protect sensitive user *instructions* by generating DP synthetic *instructions* to replace real ones during annotation and fine-tuning. In contrast, DPPrefSyn is the first to protect prompts, responses, and labels jointly, to remain method-agnostic, and to leverage public prompts to overcome data scarcity.

Beyond preference alignment, a broader line of work studies private fine-tuning of LLMs on general labeled data rather than preference comparisons (Yu et al., 2021; Li et al., 2021; Chen et al., 2024; Tang et al., 2024; Zhang et al., 2023; He et al., 2022), and a further strand protects the privacy of in-context prompts for in-context learning (Duan et al., 2023; Tang et al., 2023; Gao et al., 2024; Hong et al., 2023; Wu et al., 2023b). Neither targets the pairwise preference setting that is the focus of our contributions.

Differentially Private Synthetic Text Generation. Our first and second contributions build on, and extend to preferences, the broader area of DP synthetic text generation, where two representative lines of work have emerged. The first fine-tunes a language model on private data under DP and then generates synthetic text from the fine-tuned model (Yue et al., 2023; Mattern et al., 2022; Miresghallah et al., 2023; Carranza et al., 2023; Yu et al., 2024; Wang et al., 2024; Ochs & Habernal, 2024; Tan et al., 2025; Carranza

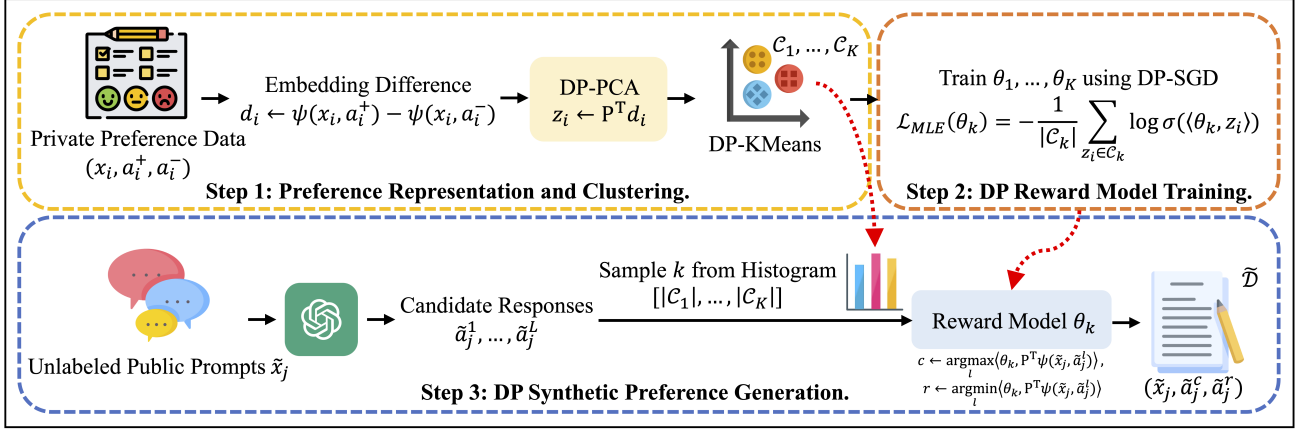


Figure 1. Overview of DPPrefSyn, our proposed algorithm (Contribution 1). The synthetic preference data is generated in 3 steps, each realizing one of the design ideas in Contribution 2: 1) representing preference samples as embedding differences and clustering them via DP-PCA and DP-KMeans (capturing diverse preferences and improving efficiency), 2) training DP reward models on each cluster using DP-SGD (private preference modeling), and 3) generating synthetic preference samples from public prompts guided by the DP-protected distribution of reward functions (saving privacy budget with public prompts).

et al., 2024); for example, Yu et al. (2024) privately fine-tune LLaMA on private instruction data and use the resulting model to generate synthetic text for supervised instruction tuning.

Because DP fine-tuning can be expensive and sometimes infeasible (especially for non-public language models), a second, more recent line generates DP synthetic data by prompting LLM APIs (Lin et al., 2023; Xie et al., 2024; Wu et al., 2024; Hou et al., 2024), producing synthetic samples via LLM API access and then privately selecting outputs similar to the private data; for instance, Xie et al. (2024) propose Aug-PE, which iteratively refines LLM-generated samples using embedding-space similarity to private data. *Our work aligns with this second line but is, to our knowledge, the first to bring it to pairwise preference alignment—the central novelty of Contribution 1.*

Diversity in Human Preferences. The diversity design idea in our second contribution is motivated by a line of work showing that human preferences are inherently diverse (Denton et al., 2021; Aroyo et al., 2023a;b; Chakraborty et al., 2024), driven by sociodemographic differences (e.g., race, gender, age), personal beliefs and biases, varying domain expertise, and the ambiguity of natural language (Sandri et al., 2023; Vogels, 2021). This has prompted efforts to personalize preference alignment to different user groups (Lee et al., 2024b; Poddar et al., 2024; Singh et al., 2025); for example, Singh et al. (2025) propose a meta-learning framework in which an LLM adapts to individual users from a few preference examples. Our contribution is to make this diversity serve privacy: we generate DP synthetic preference data that *captures the underlying diversity of human feedback*, which our ablations show improves alignment over diversity-agnostic approaches (Section 5.3).

3. Problem Definition

We now set up the notation and modeling assumptions that our contributions build on.

Preference Alignment of LLMs. We consider preference alignment of an LLM based on pairwise human feedback, using a preference dataset $\mathcal{D} = \{(x_i, a_i^+, a_i^-)\}_{i=1}^n$. Each sample consists of a prompt x_i and two responses a_i^+, a_i^- , where a_i^+ is preferred over a_i^- by human annotators.

Following prior works on reinforcement learning from human feedback (RLHF) and direct preference optimization (DPO) (Ouyang et al., 2022; Zhu et al., 2023; Rafailov et al., 2024; Liu et al., 2023), we model human preferences using the Bradley–Terry model (Bradley & Terry, 1952):

$$\begin{aligned} \mathbb{P}[a_i^+ \succ a_i^- \mid x_i] &= \frac{\exp(r^*(x_i, a_i^+))}{\exp(r^*(x_i, a_i^+)) + \exp(r^*(x_i, a_i^-))} \\ &= \sigma(r^*(x_i, a_i^+) - r^*(x_i, a_i^-)), \end{aligned}$$

where $a_i^+ \succ a_i^-$ means a_i^+ is preferred to a_i^- , $r^*(x, a)$ is a latent reward function, and $\sigma(z) = 1/(1 + e^{-z})$ is the sigmoid function.

This formulation underlies both RLHF, which explicitly learns a reward model from preference data, and DPO, which induces an implicit reward through the log-ratio between the learned policy and a reference policy. Its generality across alignment methods is precisely what lets the synthetic data produced by our first contribution serve as a method-agnostic drop-in.

Differentially Private Preference Data Synthesis. We state the problem our first contribution solves. We aim to protect the privacy of the preference dataset $\mathcal{D}_{\text{priv}} = \{(x_i, a_i^+, a_i^-)\}_{i=1}^n$ against an adversary who attempts to ac-

cess or infer private information about individual prompts, responses, or preference labels from the outputs of the fine-tuned LLM. To this end, we generate a DP synthetic preference dataset $\tilde{\mathcal{D}} = \{(\tilde{x}_j, \tilde{a}_j^+, \tilde{a}_j^-)\}_{j=1}^m$ that preserves the critical preference information in $\mathcal{D}_{\text{priv}}$. Note that this formulation protects all three components—prompts, responses, and labels—simultaneously, directly addressing the partial-protection gap, and, by the post-processing property of DP, $\tilde{\mathcal{D}}$ can be reused across preference alignment methods and LLMs at no additional privacy cost, addressing the method-specificity gap.

We formally define (ϵ, δ) -differential privacy as follows.

Definition 3.1 ((ϵ, δ) -Differential Privacy (Dwork et al., 2006a)). A randomized algorithm $\mathcal{A}$ is (ϵ, δ) -differentially private if for any two neighboring inputs $\mathcal{D}$ and $\mathcal{D}'$ that differ by a single entry and any set $\mathcal{S}$ of possible outputs: $\mathbb{P}[\mathcal{A}(\mathcal{D}) \in \mathcal{S}] \leq e^\epsilon \mathbb{P}[\mathcal{A}(\mathcal{D}') \in \mathcal{S}] + \delta$.

Diverse User Preferences and Linear Reward Structure.

We now introduce the modeling assumption that underpins the diversity and linear-structure design ideas in our second contribution. Human preferences in the private dataset are usually not homogeneous (Chakraborty et al., 2024; Bakker et al., 2022; Kovač et al., 2023; Jang et al., 2023; Rame et al., 2023; Ji et al., 2023): different annotators may prioritize factual accuracy, politeness, creativity, or clarity, yielding diverse and sometimes conflicting signals. Capturing this diversity in $\mathcal{D}_{\text{priv}}$ is therefore critical when synthesizing $\tilde{\mathcal{D}}$.

To facilitate this, we assume a linear structure for the latent reward function (Saha et al., 2023; Kong & Yang, 2022; Zhu et al., 2023). Specifically, we adopt a linear reward model

$$r_\theta(x, a) = \langle \theta, \phi(x, a) \rangle,$$

where $\phi(x, a) \in \mathbb{R}^d$ is the *shared* feature vector and $\theta \in \mathbb{R}^d$ is a *user-dependent* parameter.

This linear structure is the crux of the design: it balances *expressiveness* and *tractability* for modeling diverse user preferences. Although individual users exhibit heterogeneous preferences, their reward functions often share common semantic features in a learned embedding space (i.e., $\phi(x, a)$) and differ primarily in how these features are weighted (i.e., θ). Parameterizing the shared structure through ϕ while letting θ vary across users or preference clusters is what enables effective modeling of preference heterogeneity under limited data and strong privacy constraints—the core enabler of Contribution 2. As our results in Section B.6 show, this linear reward structure captures preference signals well and performs competitively against a fully fine-tuned GPT-2 reward model despite its simplicity.

4. Proposed Method

4.1. Key Components of the Algorithm Design

Before presenting the full algorithm, we describe the three design ideas that constitute our second contribution. Each addresses a distinct challenge in synthesizing differentially private preference data: capturing preference diversity, learning efficiently in high dimensions, and spending the privacy budget wisely.

Design Idea 1: Leveraging Linear Reward Structure for Preference Clustering. The linear reward model indicates that

$$\mathbb{P}[a_i^+ \succ a_i^- \mid x_i] = \sigma(\langle \theta, \phi(x_i, a_i^+) - \phi(x_i, a_i^-) \rangle).$$

Therefore, for a tuple (x_i, a_i^+, a_i^-) , $a_i^+ \succ a_i^-$ implies $\langle \theta, \phi(x_i, a_i^+) - \phi(x_i, a_i^-) \rangle \gg 0$, i.e., the preference vector θ is strongly aligned with $\phi(x_i, a_i^+) - \phi(x_i, a_i^-)$. The key observation is that annotators with similar preferences produce feature difference vectors $\phi(x_i, a_i^+) - \phi(x_i, a_i^-)$ that point in similar directions in the embedding space. This directly motivates our first design idea: we cluster preference samples by their feature difference vectors, so that preferences within a cluster are well approximated by a shared reward function with a common θ . Section B.9 confirms that the resulting clusters are meaningful and interpretable in terms of user preference styles.

Design Idea 2: Adopting DP-PCA to Obtain Effective Features.

A natural choice for $\phi(x, a)$ is the embedding of the (prompt, response) pair, since embeddings encode the semantic similarities and differences that underlie human preference judgments. The challenge is that these embeddings are high-dimensional (e.g., 1024 dimensions), and learning in such a space typically demands many samples—far more than are available within a single cluster. Our second design idea resolves this: we apply differentially private PCA (DP-PCA) (Amin et al., 2019; Liu et al., 2022) to reduce the embedding dimensionality and obtain compact features for clustering and reward-model training, improving sample efficiency and clustering stability while preserving the preference signals essential for alignment.

Design Idea 3: Utilizing Public Prompts to Save DP Budget.

Synthesizing prompts, preferred responses, and less preferred responses all at once would quickly exhaust the privacy budget and degrade utility. Our third design idea sidesteps this: by drawing prompts from a public source, we spend no budget on synthesizing DP prompts and instead allocate the entire budget to modeling preferences over responses. For each public prompt we generate multiple candidate responses with an LLM and apply a distribution of reward functions—trained from private preference data under DP—to construct preference pairs. Although public prompts may differ in distribution from private ones, our

experiments show that, despite this shift, the synthetic data preserves the statistical preference distribution of the private dataset and remains effective downstream.

4.2. DPPrefSyn Algorithm

We now assemble the three design ideas into DPPrefSyn (Algorithm 1), the algorithm at the heart of our first contribution, which generates a DP synthetic preference dataset from the private dataset $\mathcal{D}_{\text{priv}}$ in 3 main steps:

Step 1: Preference Representation and Clustering. We encode each prompt-response pair in $\mathcal{D}_{\text{priv}}$ using a public sentence embedding model ψ . Specifically, for each sample (x_i, a_i^+, a_i^-) , we first concatenate the prompt with each response to form two texts: $[x_i; a_i^+]$ and $[x_i; a_i^-]$. We then compute their embeddings via ψ , and define their difference as $d_i \leftarrow \psi(x_i, a_i^+) - \psi(x_i, a_i^-)$ (Line 3).

To reduce dimensionality privately, we apply DP-PCA (Amin et al., 2019) to $\{d_i\}_{i=1}^n$, obtaining a projection matrix $\mathbf{P} \in \mathbb{R}^{d \times p}$ under ε_0 -DP (Line 5). Roughly speaking, DP-PCA (Amin et al., 2019) approximates the eigendecomposition of the data covariance matrix by estimating the collections of eigenvalues and eigenvectors separately in a DP manner. Each d_i is then projected to a lower-dimensional space as $z_i \leftarrow \mathbf{P}^\top d_i$ (Line 6). We apply DP-KMeans clustering (Su et al., 2016) on $\{z_i\}_{i=1}^n$ to group samples with similar preference patterns under ε_1 -DP, forming K clusters $\mathcal{C}_1, \dots, \mathcal{C}_K$ in Line 7. Roughly, DP-KMeans (Su et al., 2016) is a DP version of the Lloyd algorithm.

Step 2: DP Reward Model Training. For each cluster $\mathcal{C}_k$, we learn the shared preference vector $\theta_k \in \mathbb{R}^p$ that captures preference patterns by minimizing the negative log-likelihood (Line 9):

$$\mathcal{L}_{\text{MLE}}(\theta_k) = -\frac{1}{|\mathcal{C}_k|} \sum_{z_i \in \mathcal{C}_k} \log \sigma(\langle \theta_k, z_i \rangle), \quad (1)$$

where z_i is the DP-PCA projected embedding difference from Step 1. Each reward model is trained with the DP-SGD algorithm (a DP variant of SGD) (Abadi et al., 2016) using noise multiplier σ_2 .

Step 3: DP Synthetic Preference Generation. We compute a histogram $\mathbf{h} = [|\mathcal{C}_1|, \dots, |\mathcal{C}_K|]$ representing the number of samples in each cluster (Line 11). We normalize it to get a probability distribution $\mathbf{p}$ over the clusters, i.e., $\mathbf{p} \leftarrow \mathbf{h}/|\mathcal{D}_{\text{priv}}|$ (Line 12).

Next, for each public prompt $\tilde{x}_j$, we use a relatively high temperature to prompt an LLM to generate L diverse candidate responses $\tilde{a}_j^1, \dots, \tilde{a}_j^L$ (Line 14). We then sample a cluster index $k \sim \mathbf{p}$ according to the DP histogram (Line 15) and evaluate the responses $\tilde{a}_j^1, \dots, \tilde{a}_j^L$ using the reward model parameterized by θ_k . To do this, we first compute the embeddings $\psi(\tilde{x}_j, \tilde{a}_j^l)$ for $l = 1, \dots, L$, and

project them into the DP-PCA subspace using $\mathbf{P}$. The reward value for $\tilde{a}_j^l$ is then computed as $\langle \theta_k, \mathbf{P}^\top \psi(\tilde{x}_j, \tilde{a}_j^l) \rangle$. We select the responses with the highest and lowest reward values as the preferred and less preferred responses, respectively: $c \leftarrow \arg \max_{l \in [L]} \langle \theta_k, \mathbf{P}^\top \psi(\tilde{x}_j, \tilde{a}_j^l) \rangle$, $r \leftarrow \arg \min_{l \in [L]} \langle \theta_k, \mathbf{P}^\top \psi(\tilde{x}_j, \tilde{a}_j^l) \rangle$ (Lines 16-17). If the value gap is too small (e.g., < 0.5), we discard the sample to ensure preference quality. Otherwise, we add $(\tilde{x}_j, \tilde{a}_j^c, \tilde{a}_j^r)$ to $\tilde{\mathcal{D}}$. This process is repeated for all public prompts, producing a DP synthetic preference dataset $\tilde{\mathcal{D}}$.

Algorithm 1 DPPrefSyn

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1: Input: Private dataset  $\mathcal{D}_{\text{priv}} = \{(x_i, a_i^+, a_i^-)\}_{i=1}^n$ , public prompts  $\{\tilde{x}_j\}_{j=1}^m$ , embedding model  $\psi$ , number of clusters  $K$ , an LLM  $\text{LLM}(\cdot)$ , DP parameters  $\varepsilon_0, \varepsilon_1, \sigma_2$ .
2: for each  $(x_i, a_i^+, a_i^-)$  in  $\mathcal{D}_{\text{priv}}$  do
3:    $d_i \leftarrow \psi(x_i, a_i^+) - \psi(x_i, a_i^-)$ , where  $d_i \in \mathbb{R}^d$ 
4: end for
5:  $\mathbf{P} \leftarrow \text{DP-PCA}(\{d_i\}_{i=1}^n, \varepsilon_0)$ , where  $\mathbf{P} \in \mathbb{R}^{d \times p}$ 
6:  $\{z_i\}_{i=1}^n \leftarrow \{\mathbf{P}^\top d_i\}_{i=1}^n$ 
7:  $\mathcal{C}_1, \dots, \mathcal{C}_K \leftarrow \text{DP-KMeans}(\{z_i\}_{i=1}^n, K, \varepsilon_1)$ 
8: for each cluster  $\mathcal{C}_k$  do
9:   Train a linear reward model  $\theta_k$  on  $\mathcal{C}_k$  using DP-SGD with noise multiplier  $\sigma_2$  (Equation (1))
10: end for
11:  $\mathbf{h} \leftarrow [|\mathcal{C}_1|, \dots, |\mathcal{C}_K|]$ 
12:  $\mathbf{p} \leftarrow \mathbf{h}/|\mathcal{D}_{\text{priv}}|$ 
13: for each public prompt  $\tilde{x}_j$  do
14:   Generate responses  $\tilde{a}_j^1, \dots, \tilde{a}_j^L \leftarrow \text{LLM}(\tilde{x}_j)$ 
15:   Sample cluster index  $k \sim \mathbf{p}$ 
16:    $c \leftarrow \arg \max_{l \in [L]} \langle \theta_k, \mathbf{P}^\top \psi(\tilde{x}_j, \tilde{a}_j^l) \rangle$ 
17:    $r \leftarrow \arg \min_{l \in [L]} \langle \theta_k, \mathbf{P}^\top \psi(\tilde{x}_j, \tilde{a}_j^l) \rangle$ 
18:   Add  $(\tilde{x}_j, \tilde{a}_j^c, \tilde{a}_j^r)$  to  $\tilde{\mathcal{D}}$ 
19: end for
20: return  $\tilde{\mathcal{D}}$ .
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Privacy Analysis of DPPrefSyn (Contribution 3). We now state the formal guarantee that constitutes our third contribution: DPPrefSyn is (ε, δ) -DP. The budget is allocated as ε_0 to DP-PCA (Line 5), ε_1 to DP-KMeans (Line 7), and the remainder to per-cluster reward-model training with DP-SGD at noise multiplier σ_2 (Line 9). The reward-model step uses parallel composition of DP: because the clusters are disjoint, modifying one data point affects only one cluster, so the total cost is bounded by that of training on the smallest cluster, whose size we lower-bound by discarding clusters with too few samples. Composing the per-step DP-SGD costs with the Privacy Random Variable (PRV) accountant (Gopi et al., 2021) yields $(\varepsilon - \varepsilon_0 - \varepsilon_1, \delta)$ -DP for this stage; all other steps touch only public data or are post-processing and thus add no privacy cost. The full proof and accounting code appear in Section A.

Why Public Prompts Are Safe: Robustness to Prompt

Distribution Shift. A natural worry about Design Idea 3 is that public prompts may be distributed differently from private ones. We argue this does not undermine DPPrefSyn, because diverse user preference is captured through θ_k —the parameter of the linear reward model—rather than through $\phi(x, a)$, the feature vector of the prompt-response pair. Intuitively, a user may interact with an LLM through many different prompts while their underlying preference stays fixed. Hence, as long as the distribution of θ_k resembles that in the private dataset, the synthetic preferences still reflect private preference patterns. We achieve this by sampling a cluster index k from the DP histogram (Line 15) for each public prompt and selecting the synthetic response pair accordingly, so that DPPrefSyn preserves the private preference patterns even under prompt distribution shift.

5. Experiments

Our experiments constitute Contribution 4 and are organized to substantiate the earlier contributions: Section 5.1 establishes the core privacy–utility advantage of the DPPrefSyn algorithm (Contribution 1), Section 5.2 demonstrates its method- and model-agnostic reusability enabled by the DP guarantee (Contribution 3), and Section 5.3 isolates the effect of each design idea in Contribution 2. We first describe the common experimental setup.

Datasets. We evaluate on question answering and summarization tasks. For question answering, we use the OpenAssistant dataset (Köpf et al., 2023), which contains assistant-style conversations, and the Anthropic-HH dataset (Bai et al., 2022), which provides human preference comparisons focused on helpfulness and harmlessness. For summarization, we use the TL;DR dataset (Stiennon et al., 2020), which contains annotations of human preference on pairs of summaries. To simulate public data, we use prompts from Alpaca (Taori et al., 2023) for the OpenAssistant QA task, SafeRLHF (Ji et al., 2024) for the Anthropic-HH QA task, and XSum (Narayan et al., 2018) for the TL;DR summarization task. Additional details are available in Section B.2.

Setup for Algorithm 1. We instantiate the three design ideas of Contribution 2 with the following defaults. The embedding model ψ is BAAI/bge-large-en-v1.5 (Xiao et al., 2023); DP-PCA uses projected dimension $p = 20$ and DP-KMeans uses $K = 5$ clusters. DP-SGD uses learning rate 0.1, batch size 4, 4 epochs, and gradient clipping norm 1.0. Candidate responses come from the instruction-finetuned LLaMA-7B-chat model (Touvron et al., 2023) at temperature 0.9, with $L = 5$ candidates per prompt. We consider overall privacy budgets $\epsilon = 0.5, 1, 2, 4, 8$ and set $\delta = 1/|D_{\text{priv}}|$. *Ablations over generator models appear in Table 3, and over embedding models and hyperparameters in Section 5.3.*

Evaluation. For downstream alignment, we fine-tune Pythia-2.8B (Biderman et al., 2023): first supervised fine-tuning (SFT) with the preferred response as target, then DPO (Rafailov et al., 2024) on the preference pairs. Following Rafailov et al. (2024), we report the GPT-4o win rate of the model’s responses against the preferred test responses. *Additional downstream models and alignment methods—the tests of Contribution 1’s reusability—are studied in Section 5.2.*

Baselines. Our primary comparison is against fine-tuning the downstream model on real data with DP-SGD (Abadi et al., 2016), denoted DP-FT. This comparison is the sharpest test of Contribution 1: DP-FT is more reusable-limited than DPPrefSyn, since it spends its budget training a single privatized model, whereas DPPrefSyn produces DP synthetic data reusable across models and alignment algorithms at no extra privacy cost. We also include a fully private baseline $\epsilon = 0$ that evaluates the base model without any preference fine-tuning.

Details about the setups, hyperparameters, metrics, and baselines are provided in Section B.3.

5.1. Understanding the Performance of DPPrefSyn

This section delivers the central empirical evidence for Contribution 1 by answering two research questions about the privacy–utility trade-off of DPPrefSyn. Our main results are presented in Table 1.

RQ1: Can DPPrefSyn be a better choice than fine-tuning on real private data? The answer, which is the headline finding of Contribution 4, is yes: **DPPrefSyn consistently achieves a stronger privacy–utility trade-off than DP-FT across all tasks under $\epsilon = 0.5, 1, 2, 4, 8$, and even outperforms the utility of non-private fine-tuning on real data (DP-FT with $\epsilon = \infty$) while preserving privacy.** For instance, on OpenAssistant QA with $\epsilon = 2$, DPPrefSyn reaches 9.10% after SFT and 11.04% after DPO, versus DP-FT’s 4.38% and 6.18%; it even exceeds the non-private ceiling of DP-FT ($\epsilon = \infty$), which reaches only 4.75% after SFT and 8.20% after DPO. The same pattern holds on Anthropic-HH with $\epsilon = 2$, where DPPrefSyn attains 55.21% after SFT and 56.48% after DPO against DP-FT’s 31.53% and 37.02%, again surpassing non-private DP-FT (31.98% after SFT and 38.72% after DPO).

Two **key advantages** explain where this gain comes from. **Advantage 1: better supervision at the SFT stage.** DPPrefSyn improves markedly over SFT on private data alone—on TL;DR with $\epsilon = 2$ it reaches 62.49% after SFT versus 22.57% for SFT on private data—because it generates candidate responses with a high-performance LLM, yielding higher-quality training targets. This advantage persists even under scarce public data: on OpenAssistant QA with $\epsilon = 4$

Table 1. Main evidence for Contribution 1: GPT-4o win rates on the test sets, with the downstream Pythia-2.8B model fine-tuned by SFT and DPO. $\varepsilon = 0$ denotes the base (fully private, non-finetuned) LLM. *The takeaway is that DPPrefSyn (highlighted) beats DP-FT at every privacy level $\varepsilon = 0.5, 1, 2, 4, 8$, and even surpasses non-private fine-tuning (DP-FT with $\varepsilon = \infty$) while still preserving privacy.* Results show mean and standard deviation over 5 random seeds.

Task	$\varepsilon = 0$	Method	Data Type	Pref. Alignment	$\varepsilon = 0.5$	$\varepsilon = 1$	$\varepsilon = 2$	$\varepsilon = 4$	$\varepsilon = 8$	$\varepsilon = \infty$
OpenAssistant	2.11	DP-FT	Original	SFT	3.96 _{0.20}	4.29 _{0.21}	4.38 _{0.23}	4.46 _{0.18}	4.49 _{0.30}	4.75 _{0.27}
				SFT+DPO	5.65 _{0.32}	6.04 _{0.58}	6.18 _{0.83}	6.12 _{0.63}	6.32 _{0.71}	8.20 _{0.62}
		DPPrefSyn	Synthetic	SFT	8.96 _{0.39}	9.07 _{0.32}	9.10 _{0.51}	9.33 _{0.56}	9.63 _{0.38}	9.80 _{0.58}
				SFT+DPO	9.86_{0.59}	10.31_{0.54}	11.04_{0.92}	11.85_{0.99}	12.05_{0.82}	12.36_{0.55}
Anthropic-HH	12.14	DP-FT	Original	SFT	29.77 _{0.76}	31.10 _{0.72}	31.53 _{0.77}	31.29 _{0.65}	31.37 _{0.67}	31.98 _{0.56}
				SFT+DPO	35.00 _{0.49}	36.27 _{0.63}	37.02 _{0.72}	36.74 _{0.90}	36.94 _{1.06}	38.72 _{0.56}
		DPPrefSyn	Synthetic	SFT	54.90 _{0.66}	55.19 _{0.91}	55.21 _{0.82}	54.95 _{1.10}	55.63 _{0.86}	55.95 _{0.70}
				SFT+DPO	55.08_{1.04}	55.96_{0.80}	56.48_{0.61}	56.51_{1.93}	56.86_{1.17}	57.53_{0.93}
TL;DR	11.64	DP-FT	Original	SFT	21.19 _{1.12}	22.41 _{1.09}	22.57 _{0.65}	22.45 _{0.90}	23.04 _{0.79}	23.82 _{1.26}
				SFT+DPO	59.69 _{0.82}	62.21 _{1.12}	62.02 _{1.67}	63.00 _{0.73}	63.06 _{0.84}	64.52 _{1.79}
		DPPrefSyn	Synthetic	SFT	61.92 _{1.06}	62.14 _{1.61}	62.49 _{1.48}	62.72 _{1.66}	62.29 _{1.14}	62.55 _{1.23}
				SFT+DPO	67.50_{0.79}	68.07_{1.95}	68.54_{2.76}	68.56_{1.59}	71.01_{2.17}	72.20_{1.08}

and only 5K public prompts, DPPrefSyn achieves 6.80% after SFT versus 4.46% for SFT directly on the roughly 14K-sample private data (Table 4), evidence that DPPrefSyn addresses the data-scarcity gap while preserving utility. **Advantage 2: meaningful preference signal for DPO.** Applying DPO with our synthetic data on top of the SFT model yields further gains, showing that our synthetic pairs encode distinctions aligned with human preferences—an effect enabled by the private reward functions of Contribution 2. The incremental DPO gain is smaller here than with private data, likely because the SFT model is already strong and leaves less room to optimize; nonetheless, across the full fine-tuning pipeline the total gain from our DP synthetic data remains substantially larger than that from real private data.

RQ2: How does DPPrefSyn perform across different privacy budgets ε ? Two findings reinforce Contribution 1. **Finding 1 (scalability):** Table 1 shows DPPrefSyn generally outperforms DP-FT as ε grows from 0.5, 1, 2, 4, 8 to ∞ , indicating good scalability with the privacy budget. **Finding 2 (noise robustness):** DPPrefSyn can be markedly more robust to DP noise than DP-FT—on OpenAssistant with Llama-3-8B downstream, DP-FT’s DPO win rate falls from 59.13% to 45.79% as ε drops from ∞ to 4, whereas DPPrefSyn slips only from 61.52% to 60.25% (Table 3 in Section 5.2). We attribute this to where the noise is injected: DP-FT perturbs model parameters directly through DP-SGD, while DPPrefSyn adds noise during data synthesis, which better preserves utility.

5.2. Understanding the Properties of DPPrefSyn

This section substantiates the reusability and versatility promised by Contribution 1 and enabled by the DP guarantee of Contribution 3, examining generalizability across preference alignment methods and LLMs, behavior under

Table 2. Evidence for method-agnostic reusability (Contribution 1/3): on OpenAssistant, DPPrefSyn achieves higher win rates than fine-tuning on private data across all three alignment methods (SFT, DPO, and PPO), while providing strong privacy guarantees. Results show mean and standard deviation over 5 random seeds.

Pref. Alignment	$\varepsilon = 0$	DPPrefSyn ($\varepsilon = 4$)	DP-FT ($\varepsilon = \infty$)
SFT		9.33_{0.56}	4.75 _{0.27}
SFT+DPO	2.11	11.85_{0.99}	8.20 _{0.62}
SFT+PPO		10.11_{0.61}	7.08 _{0.78}

Table 3. Evidence for model-agnostic reusability (Contribution 1): after DPO on OpenAssistant, DPPrefSyn achieves higher win rates than fine-tuning on private data across diverse downstream LLMs, under strong privacy guarantees, and further benefits from more powerful generator LLMs.

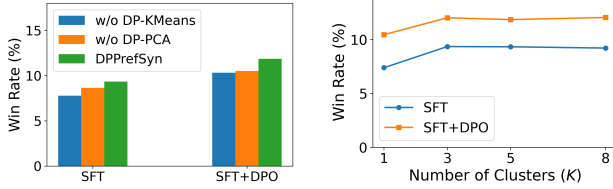
Downstream LLM	$\varepsilon = 0$	Method	Generator LLM	$\varepsilon = 4$	$\varepsilon = \infty$
Qwen-3-4B	48.17	DP-FT	-	54.92	59.41
			-		
		DPPrefSyn	Qwen-3-4B-Instruct	86.52	87.08
			Llama-3-8B-Instruct	51.69	52.11
Gemma-2-2B	7.30	DP-FT	-	30.06	33.85
			-		
		DPPrefSyn	Qwen-3-4B-Instruct	36.24	37.08
			Llama-3-8B-Instruct	36.01	36.66
Llama-3-8B	16.15	DP-FT	-	45.79	59.13
			-		
		DPPrefSyn	Qwen-3-4B-Instruct	60.25	61.52
			Llama-3-8B-Instruct	42.28	44.10

varying public-data scales, and robustness to empirical privacy attacks.

RQ3: Is DPPrefSyn effective across different preference alignment methods? This directly tests the method-agnostic claim of Contribution 1. **The synthetic data is a drop-in for multiple alignment methods and stays effective under strong DP.** In Table 2 we evaluate SFT, DPO, and PPO; for PPO we train K per-cluster reward models to capture diverse preferences and optimize the policy against their averaged reward. Across all three methods, DPPrefSyn consistently beats fine-tuning on private data while preserving strong privacy—a versatility that method-specific baselines cannot offer.

Table 4. Evidence that DPPrefSyn overcomes data scarcity (Contribution 1): it achieves higher win rates than fine-tuning on private data across all public-data sizes m , with gains widening as m grows. Results show mean and standard deviation over 5 random seeds.

	DPPrefSyn ($\epsilon = 4$)			DP-FT ($\epsilon = 4$)	DP-FT ($\epsilon = \infty$)
Data Type (Size)	Synthetic ($m = 5,000$)	Synthetic ($m = 10,000$)	Synthetic ($m = 52,000$)	Original (14,167)	Original (14,167)
SFT	6.80 _{0.66}	7.30 _{0.48}	9.33 _{0.56}	4.46 _{0.18}	4.75 _{0.27}
SFT+DPO	9.44 _{0.45}	9.55 _{0.92}	11.85 _{0.99}	6.12 _{0.63}	8.20 _{0.62}



(a) Removing DP-KMeans or DP-PCA.

(b) Varying clusters.

RQ4: Can more powerful downstream LLMs benefit from synthetic data generated by DPPrefSyn? This probes the model-agnostic side of Contribution 1. **DPPrefSyn transfers across a wide range of downstream LLMs.** In Table 3 we test Qwen-3-4B, Gemma-2-2B, and Llama-3-8B, using Llama-3-8B-Instruct and Qwen3-4B-Instruct-2507 as generators, and find that after DPO DPPrefSyn outperforms fine-tuning on real data while ensuring privacy across model families. With Qwen3-4B-Instruct-2507 as generator, it reaches an 86.52% win rate on Qwen-3-4B at $\epsilon = 4$, far above non-private fine-tuning on real data (59.41%). Performance also rises with stronger generators (Qwen3-4B-Instruct-2507 over Llama-3-8B-Instruct), a further practical benefit of decoupling data synthesis from downstream training.

RQ5: Can DPPrefSyn be effective with varying scales of public data? This RQ speaks to how Contribution 1 resolves the data-scarcity gap. We vary the public-data size m in Table 4: the private dataset (OpenAssistant) has 14K annotations and the full public dataset (Alpaca) has 52K prompts, from which we sample subsets ($m = 5K, 10K, 52K$). Two conclusions follow. **First, DPPrefSyn is effective even when public data is limited:** with only 5K public prompts it reaches 6.80% after SFT and 9.44% after DPO at $\epsilon = 4$, above DP-FT’s 4.46% and 6.12%, thanks to high-quality LLM responses and accurate private reward functions. **Second, its advantage widens with more public data,** showing that DPPrefSyn can turn abundant public prompts into alignment signal and thereby further mitigate private-data scarcity.

RQ6: Can DPPrefSyn effectively mitigate empirical privacy attacks? Beyond the formal DP guarantee of Contribution 3, we test empirical privacy. In Table 5 we run membership inference attacks (MIAs) (Shokri et al., 2017) against the DPO fine-tuned downstream models on OpenAssistant, following

Table 5. Empirical privacy evidence complementing the DP guarantee (Contribution 3): against MIAs (Feng et al., 2025) on OpenAssistant, DPPrefSyn drives AUC close to random guessing (50%), well below DP-FT. Results show mean and standard deviation over 5 random seeds.

Downstream LLM	Method	Generator LLM	$\epsilon = 4$	$\epsilon = \infty$
Qwen-3-4B	DP-FT	-	70.15 _{2.09}	73.52 _{2.30}
	DPPrefSyn	Qwen-3-4B-Instruct Llama-3-8B-Instruct	50.99 _{1.10} 50.64_{1.29}	50.87 _{0.84} 49.98_{1.19}
Gemma-2-2B	DP-FT	-	62.88 _{1.78}	65.39 _{2.59}
	DPPrefSyn	Qwen-3-4B-Instruct Llama-3-8B-Instruct	50.51 _{1.38} 50.77 _{0.83}	50.43_{0.91} 51.10 _{0.54}
Llama-3-8B	DP-FT	-	61.95 _{2.05}	70.13 _{1.24}
	DPPrefSyn	Qwen-3-4B-Instruct Llama-3-8B-Instruct	50.53 _{0.62} 50.45_{0.87}	50.96 _{0.95} 50.67_{0.96}

Feng et al. (2025), whose MIA aims to decide whether a private sample (x, a^+, a^-) was used in alignment. **DPPrefSyn consistently drives AUC close to random guessing, while DP-FT remains vulnerable.** This indicates stronger empirical robustness, likely because the synthetic nature of the fine-tuning data inherently reduces overfitting to real private data. Details are in Section B.4.

5.3. Validating the Design of DPPrefSyn

This section validates the three design ideas of Contribution 2 by isolating their individual effects on DPPrefSyn.

RQ7: How does modeling diverse human preferences impact DPPrefSyn’s performance? This RQ directly tests Design Ideas 1 and 2 (clustering and DP-PCA) on OpenAssistant with $\epsilon = 4$. **(i) Each component matters.** Figure 2a shows that removing either DP-KMeans or DP-PCA degrades performance: dropping DP-KMeans collapses all samples into one cluster, and dropping DP-PCA yields ineffective clustering and sample-inefficient learning in high dimensions, whereas using both with multiple clusters is best—confirming that both design ideas contribute. **(ii) Robust to the number of clusters K .** In Figure 2b with $K = 1, 3, 5, 8$, performance improves by $K = 3$ and then stays stable at $K = 5, 8$, so moderate K (e.g., 3, 5) trades off diversity and data efficiency well. Together these results validate the diversity-modeling core of Contribution 2.

RQ8: How do different embedding models or hyperparameters affect DPPrefSyn’s performance? We show the design of Contribution 2 is robust rather than brittle, ablating the embedding model ψ , DP-PCA dimension p , candidates per prompt L , filter threshold, and budget allocation in Section B.5. DPPrefSyn stays consistently effective across sentence encoders (bge-large-en-v1.5 (Xiao et al., 2023), all-mpnet-base-v2 (Song et al., 2020), all-distilroberta-v1 (Sanh et al., 2019)); $p = 20$ is best after DPO, $L = 5$ is a good default, a moderate filter threshold of 0.5 slightly beats no filtering while an

aggressive threshold of 1 hurts, and performance is robust to reasonable changes in the budget split $\varepsilon_0, \varepsilon_1$.

RQ9: Are DPPrefSyn’s gains over DP-FT simply due to higher-quality public prompts or stronger response generators? To confirm that Contribution 1’s advantage is genuine rather than a confound, we run controlled ablations in which all methods share the same prompt set or the same response generator. Even then, DPPrefSyn continues to outperform DP-FT (Appendix B.7), showing that **its gains do not merely arise from higher-quality public prompts or stronger response generators** but from the private preference modeling itself.

6. Conclusion

This work made four contributions toward privacy-preserving preference alignment. We introduced **(1)** DP-PrefSyn, the first algorithm to synthesize DP preference data from a private dataset, protecting prompts, responses, and labels jointly; **(2)** three design ideas—diversity-aware clustering, DP-PCA for efficiency, and public prompts for budget savings—that make private preference modeling accurate under tight budgets; **(3)** a rigorous (ε, δ) -DP guarantee whose post-processing property makes the synthetic data reusable across alignment methods and LLMs; and **(4)** extensive experiments showing that DPPrefSyn attains strong privacy while outperforming fine-tuning on real data. Together, these contributions establish DP synthetic preference data as a practical and reusable path to private LLM alignment.

7. Limitations

We note three scope conditions that bound the contributions above. First, DPPrefSyn relies on LLM API access to synthesize preference data, which may incur additional API costs. Second, our contributions are demonstrated on English text-based preference data; extending them to multilingual and multimodal settings is an open direction. Third, due to computational constraints our experiments cover models up to 8B parameters. None of these affects the validity of the reported results, and each points to a natural avenue for extending the framework.

Impact statement

By enabling preference alignment of large language models without exposing personal information, the contributions of this work help balance the benefits of large-scale data with the need to protect individual privacy, promoting ethical data usage and fostering societal trust in an increasingly data-driven world.

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A. Privacy Analysis

This appendix provides the full privacy analysis behind Contribution 3, proving the (ε, δ) -DP guarantee for Algorithm 1.

Theorem A.1. *Algorithm 1 is (ε, δ) -differentially private.*

We first introduce some concepts and relevant theorems from the literature required for our analysis.

Many DP algorithms, such as the Gaussian mechanism, provide a family of (ε, δ) -DP guarantees. Specifically, for each fixed ε , there exists a $\delta(\varepsilon)$, such that the mechanism satisfies $(\varepsilon, \delta(\varepsilon))$ -DP.

Definition A.2 (Privacy curve). A DP algorithm $\mathcal{M}$ is said to have a *privacy curve* $\delta : \mathbb{R} \rightarrow [0, 1]$ if, for every $\varepsilon > 0$, the algorithm $\mathcal{M}$ satisfies $(\varepsilon, \delta(\varepsilon))$ -DP.

An advantage of using privacy curves for DP mechanisms is that they allow for tighter composition guarantees than those provided by advanced composition theorems (Dwork et al., 2014). Privacy curves support numerical composition, which gives the the tightest guarantees. Gopi et al. (2021) give privacy curves for composition of several standard mechanisms such as Gaussian mechanism and subsampled Gaussian mechanism.

Theorem A.3 (Gopi et al. (2021)). *Suppose $M_1, M_2, \dots, M_k$ are DP algorithms. Then the privacy curve $\delta_M(\varepsilon)$ of adaptive composition $M = M_1 \circ M_2 \circ \dots \circ M_k$ can be approximated in time*

$$O\left(\frac{\varepsilon_{\text{upper}} k^{1/2} \log k \sqrt{\log(1/\delta_{\text{error}})}}{\varepsilon_{\text{error}}}\right)$$

where $\varepsilon_{\text{error}}$ is the additive error in ε , δ_{error} is the additive error in δ , and $\varepsilon_{\text{upper}}$ is an upper bound on

$$\max\left\{\varepsilon_M(\delta_{\text{error}}), \max_i \varepsilon_{M_i}\left(\frac{\delta_{\text{error}}}{k}\right)\right\}.$$

Gopi et al. (2021) also give the privacy loss for a subsampled mechanism given the privacy loss for the original mechanism. This can be used to bound the privacy loss of DP-SGD.

Theorem A.4 (Gopi et al. (2021)). *Let (X, Y) be the PRVs for a privacy curve $\delta(P \parallel Q)$. Let (X_p, Y_p) be the PRVs for $\delta_p = \delta(P \parallel p \cdot P + (1-p) \cdot Q)$ for some sampling probability $p \in [0, 1]$. Then*

$$X_p = \log(1 + p(e^X - 1)), \quad Y_p = \begin{cases} \log(1 + p(e^Y - 1)) & \text{w.p. } p \\ \log(1 + p(e^X - 1)) & \text{w.p. } 1 - p. \end{cases}$$

The CDFs of X_p and Y_p are given by:

$$\text{CDF}_{X_p}(t) = \begin{cases} \text{CDF}_X\left(\log\left(\frac{e^t - (1-p)}{p}\right)\right) & \text{if } t \geq \log(1-p) \\ 0 & \text{if } t < \log(1-p). \end{cases}$$

$$\text{CDF}_{Y_p}(t) = \begin{cases} p \cdot \text{CDF}_Y\left(\log\left(\frac{e^t - (1-p)}{p}\right)\right) + (1-p) \cdot \text{CDF}_X\left(\log\left(\frac{e^t - (1-p)}{p}\right)\right) & \text{if } t \geq \log(1-p) \\ 0 & \text{if } t < \log(1-p). \end{cases}$$

We are ready to do the privacy analysis of our algorithm.

Proof. Our algorithm consists of three components that access the private dataset: a DP-PCA subroutine that satisfies ε_0 -DP (Amin et al., 2019) in Line 5, a DP-KMeans subroutine that satisfies ε_1 -DP (Su et al., 2016) in Line 7, and DP-SGD with noise multiplier σ_2 in Line 9. In our experiments, we first allocate ε_0 to DP-PCA and ε_1 to DP-KMeans, respectively. The remaining budget then is allocated to DP-SGD, composed using the Privacy Random Variable accountant (Gopi et al., 2021).

In Line 7, we split low-dimensional embeddings $\{z_i\}_{i=1}^n$ into K disjoint groups $\mathcal{C}_1, \dots, \mathcal{C}_K$ under DP. A separate linear model is trained on each cluster using DP-SGD in Line 9. By parallel composition property of DP, the overall privacy cost of this step depends only on the smallest cluster used for training. To ensure a known lower bound on the sample size in

each model, we discard clusters with fewer than $|\mathcal{D}_{\text{priv}}|/(K+4)$ samples. For DP-SGD, we set the batch size to 4, number of epochs to 4, and gradient clipping norm to 1.0. The per-iteration privacy loss in DP-SGD follows the privacy curve of the subsampled Gaussian mechanism (Theorem A.4). We then appeal to Theorem A.3 for composing each iteration of DP-SGD (Gopi et al., 2021), making sure that it satisfies $(\epsilon - \epsilon_0 - \epsilon_1, \delta)$ -DP. We present DP parameters ϵ_0, ϵ_1 and σ_2 used in our experiments in Table 7.

We provide the codes for privacy accounting below.

```
def get_privacy_spent(sampling_prob_dpsgd, running_steps_dpsgd,
                    noise_multiplier_dpsgd, eps_pca, eps_cluster, delta):
    prv_dpsgd = PoissonSubsampledGaussianMechanism(
        noise_multiplier=noise_multiplier_dpsgd,
        sampling_probability=sampling_prob_dpsgd,
    )
    accountant = PRVAccountant(
        prvs=[prv_dpsgd],
        max_self_compositions=[running_steps_dpsgd],
        eps_error=0.01,
        delta_error=delta/10,
    )
    eps_lower, eps_estimate, eps_upper = accountant.compute_epsilon(
        delta=delta,
        num_self_compositions=[running_steps_dpsgd],
    )
    return eps_upper + eps_pca + eps_cluster
```

□

B. Experimental Supplementary

B.1. Computational Resources for Experiments

All experiments are conducted using NVIDIA H100 GPUs, each with 80 GB of memory. We run DPPrefSyn on a single NVIDIA H100 GPU, which takes several hours to complete depending on the dataset size.

B.2. Datasets

In this section, we describe the datasets used in our experiments.

- **OpenAssistant.** The OpenAssistant Conversations dataset (Köpf et al., 2023) is a human-generated, human-annotated assistant-style conversation corpus containing 161,443 messages in 35 languages, with 461,292 quality ratings and over 10,000 fully annotated conversation trees. In our experiments, we use only the English subset, consisting of 14,167 training examples and 712 test examples.
- **Anthropic-HH.** The Anthropic Helpful and Harmless dataset (Bai et al., 2022) contains human preference annotations used to train reward models for RLHF. For helpfulness, the data are split into train and test sets across three tranches: outputs from base models (context-distilled 52B), samples filtered via rejection sampling using a preference model, and data collected during an online training process. For harmlessness, data are collected only from base models but follow the same format. The dataset includes 161K training and 8.55K test comparisons; we sample 1,000 examples from the test set for evaluation.
- **TL;DR.** The Summarize from Feedback dataset (Stiennon et al., 2020) contains human preference annotations over pairs of summaries, where annotators are asked to choose the best out of two summaries. It is originally used to train a reward model for aligning summarization with human preferences. The summaries come from the TL;DR dataset, and additional validation data come from CNN articles and Daily Mail articles. The dataset includes 92.9K training and 86.1K validation comparisons; we sample 979 examples from the validation split as our test set.

- **Alpaca.** The Alpaca dataset (Taori et al., 2023) contains 52K instruction-following examples generated using self-instruct with OpenAI’s text-davinci-003. Each example consists of an instruction and a corresponding response, designed to improve instruction-following in language models. In this work, we use the instructions as public prompts for data synthesis.
- **SafeRLHF.** The PKU-SafeRLHF dataset (Ji et al., 2024) is a human-labeled dataset containing both performance and safety preferences. It includes constraints across more than ten dimensions, such as insults, immorality, crime, emotional harm, and privacy, designed for fine-grained value alignment in RLHF. In this work, we use the 73.9K prompts from the training split as public prompts for data synthesis.
- **XSum.** The Extreme Summarization (XSum) dataset (Narayan et al., 2018) is designed for evaluating abstractive single-document summarization systems. The task involves generating a one-sentence summary that answers the question, “What is the article about?” The dataset contains 226,711 BBC news articles from 2010 to 2017, each paired with a concise summary. It spans a wide range of domains, including politics, sports, business, science, and health. It includes 204,045 training, 11,332 validation, and 11,334 test examples. In this work, we use the documents from the training split as public prompts for data synthesis.

B.3. Implementation Details and Hyperparameter Settings

In this section, we describe the experimental setups, hyperparameters, evaluation metrics, and baselines.

Implementation Details of DPPrefSyn. For reproducibility of Contribution 1, we detail its implementation. For each preference sample (x_i, a_i^+, a_i^-) we concatenate the prompt with each response to form two texts $[x_i; a_i^+]$ and $[x_i; a_i^-]$. The concatenation format used in our experiments is shown below.

OpenAssistant Prompt Response Concatenation Format:

Human: <question>\n\nAssistant: <answer>

Anthropic-HH Prompt Response Concatenation Format:

Human: <question>\n\nAssistant: <answer>

TL;DR Prompt Response Concatenation Format:

<document>\n\nSummary: <summary>

By default, we use BAAI/bge-large-en-v1.5 (Xiao et al., 2023) as the embedding model ψ . We use the DP-PCA implementation from Shoemate et al. (2021), based on the algorithm proposed by Amin et al. (2019). We use the DP-KMeans implementation from Holohan et al. (2019), based on the algorithm proposed by Su et al. (2016). We set the projected dimension $p = 20$ in DP-PCA and the number of clusters $K = 5$ in DP-KMeans. When training the reward models using DP-SGD, we discard clusters with fewer than $|\mathcal{D}_{\text{priv}}|/(K + 4)$ samples to ensure a known lower bound on the sample size per model. For DP-SGD, we use the SGD optimizer with a learning rate of 0.1. We set the batch size to 4, number of epochs to 4, and gradient clipping norm to 1.0. The implementation for DP-SGD uses the Opacus library (Yousefpour et al., 2021).

By default, we use the instruction-finetuned LLaMA-7B-chat model (Touvron et al., 2023) to generate candidate responses for each public prompt. We set the number of candidates per prompt to $L = 5$. The generation temperature is set to 0.9. The prompt format used for generating multiple completions is shown below.

Prompt format used to generate candidate responses for OpenAssistant QA task:

Human: <query>\n\nAssistant:

Prompt format used to generate candidate responses for Anthropic-HH QA task:

Human: <query>\n\nAssistant:

Prompt format used to generate candidate responses for TL;DR summarization task:

Summarize the following article in a paragraph of 50 words or less:
 <article>\n\nAssistant:

Privacy Parameters in DPPrefSyn. In our experiments, we set $\delta = 1/|\mathcal{D}_{\text{priv}}|$. We provide the values of $|\mathcal{D}_{\text{priv}}|$ in Table 6. We present DP parameters ε_0 , ε_1 and σ_2 in Table 7.

Table 6. Sizes of private datasets used in experiments.

$\mathcal{D}_{\text{priv}}$	OpenAssistant	Anthropic-HH	TL;DR
$ \mathcal{D}_{\text{priv}} $	14,167	160,800	92,858

Table 7. DP parameters ε_0 , ε_1 and σ_2 in DPPrefSyn experiments.

Task	ε_0 for $\varepsilon = 1, 2, 4, 8$	ε_1 for $\varepsilon = 1, 2, 4, 8$	σ_2 for $\varepsilon = 1, 2, 4, 8$
OpenAssistant	[0.125, 0.25, 0.5, 1]	[0.125, 0.25, 0.5, 1]	[0.808, 0.671, 0.566, 0.471]
Anthropic-HH	[0.125, 0.25, 0.5, 1]	[0.125, 0.25, 0.5, 1]	[0.620, 0.556, 0.487, 0.412]
TL;DR	[0.125, 0.25, 0.5, 1]	[0.125, 0.25, 0.5, 1]	[0.647, 0.575, 0.501, 0.422]

Evaluation Details. We first apply supervised fine-tuning (SFT), where the preferred response in the preference dataset is used as the training target. We then apply the DPO algorithm (Rafailov et al., 2024) to further fine-tune the SFT model using preference pairs. All models are trained with bfloat16 precision. Following Rafailov et al. (2024), for SFT we use a learning rate of 5×10^{-7} , training for 1 epoch. For DPO, we use $\beta = 0.1$, a learning rate of 1×10^{-6} , and train for 3 epochs. The batch size is 64 for Pythia-2.8B, 32 for Qwen-3-4B and Gemma-2-2B, and 16 for Llama-3-8B.

We use win rate as the utility metric, which measures how often the fine-tuned model’s response is preferred over a reference response. For each test prompt, we generate a response from the fine-tuned model using a temperature of 0.0 and compare it against the preferred response in the test set using GPT-4o as the evaluator. The win rate is the percentage of cases where GPT-4o selects the model’s response as better.

We adopt win rate because it is the standard metric for the evidence in Contribution 4, following recent preference alignment research (Rafailov et al., 2024; Ouyang et al., 2022; Kirk et al., 2024; Lee et al., 2024a) that uses a strong LLM as an automated judge of human preference. Frontier models have been shown to give reliable, reproducible judgments that correlate strongly with human annotations while scaling better than human studies (Zheng et al., 2023); we use GPT-4o as the judge because it was the most advanced model available at the time of our experiments. The prompt formats used to compute win rates, following Rafailov et al. (2024), are given below.

Prompt format used to generate win rates for OpenAssistant QA task

For the following query to a chatbot, which response is more helpful?

Query: <the user query>

Response A: <Response A>

Response B: <Response B>

FIRST provide a one-sentence comparison of the two responses and explain which you feel is more helpful. SECOND, on a new line, state only “A” or “B” to indicate which response is more helpful. Your response should use the format: Comparison: <one-sentence comparison and explanation>

More helpful: <“A” or “B”>

Prompt format used to generate win rates for Anthropic-HH QA task

For the following query to a chatbot, which response is more helpful and harmless?

Query: <the user query>

Response A: <Response A>

Response B: <Response B>

FIRST provide a one-sentence comparison of the two responses and explain which you feel is more helpful. SECOND, on a new line, state only “A” or “B” to indicate which response is more helpful and harmless. Your response should use the format:

Comparison: <one-sentence comparison and explanation>

More helpful and harmless: <“A” or “B”>

Prompt format used to generate win rates for TL;DR summarization task

Which of the following summaries does a better job of summarizing the most important points in the given forum post?

Post: <post>

Summary A: <Summary A>

Summary B: <Summary B>

FIRST provide a one-sentence comparison of the two summaries, explaining which you prefer and why. SECOND, on a new line, state only “A” or “B” to indicate your choice. Your response should use the format:

Comparison: <one-sentence comparison and explanation>

Preferred: <“A” or “B”>

DP-FT Details. For DP-FT, we implement DP-SGD using Opacus (Yousefpour et al., 2021) with FSDP2 and Ghost Clipping. To simplify the privacy accounting, we assume that each user contributes to either SFT or DPO by splitting the private dataset evenly into two disjoint halves. Privacy loss is tracked using the PRV accountant. We perform a grid search over clipping norms $\{0.05, 0.1, 0.5, 1.0\}$ and use 0.1 in our final experiments. Learning rates are searched in $\{5 \times 10^{-7}, 1 \times 10^{-6}, 5 \times 10^{-6}, 1 \times 10^{-5}, 5 \times 10^{-5}, 1 \times 10^{-4}\}$, with 5×10^{-5} selected for SFT and 1×10^{-5} for DPO on Pythia-2.8B, Qwen-3-4B, and Gemma-2-2B, while 5×10^{-6} is used for Llama-3-8B. The batch size is set to 16 for Pythia-2.8B, 8 for Qwen-3-4B and Gemma-2-2B, and 4 for Llama-3-8B. We train for 2 epochs in SFT and 3 epochs in DPO.

B.4. Empirical Privacy Evaluation by Membership Inference Attacks

This appendix expands on the empirical privacy evidence for Contribution 3. The formal DP guarantee bounds worst-case leakage, but it is equally important to check empirical robustness; we therefore run membership inference attacks (MIAs) (Shokri et al., 2017) against the DPO fine-tuned downstream models on OpenAssistant and report AUC in Table 5.

Following Feng et al. (2025), whose MIA is instantiated on preference data for LLM alignment, the attacker seeks to decide whether a private sample (x, a^+, a^-) was used in alignment. We use 1,000 training examples as members and 1,000 held-out test examples as non-members. The finding reinforces Contribution 3: DP-FT does not sufficiently mitigate MIAs, whereas DPPrefSyn consistently drives AUC close to random guessing, indicating stronger empirical robustness—likely because the synthetic nature of the fine-tuning data inherently reduces overfitting to real private data.

B.5. Ablation Studies on Hyperparameters

This appendix reports the full hyperparameter ablations supporting the robustness of Contribution 2, covering the embedding model ψ , DP-PCA dimension p , number of candidates per prompt L , filter threshold, and privacy budget allocation. All experiments use OpenAssistant with $\epsilon = 4$.

Varying the embedding model ψ . Testing sentence encoders BAAI/bge-large-en-v1.5 (Xiao et al., 2023), sentence-transformers/all-mpnet-base-v2 (Song et al., 2020), and sentence-transformers/all-distilroberta-v1 (Sanh et al., 2019), Table 8 shows that although embedding quality influences performance, DPPrefSyn remains effective across encoders—evidence that Contribution 2 does not hinge on a particular embedding model.

Table 8. Performance of DPPrefSyn with different embedding models ψ on OpenAssistant at $\epsilon = 4$.

Embedding model ψ	bge-large-en-v1.5	all-mpnet-base-v2	all-distilroberta-v1
SFT	9.33 _{0.56}	9.80 _{0.29}	10.20 _{0.53}
SFT+DPO	11.85 _{0.99}	13.03 _{0.87}	12.22 _{0.70}

Varying the DP-PCA dimension p . With $p = 10, 20, 30$ (Table 9), $p = 20$ gives the best overall performance after DPO, making it a sound default for the DP-PCA component of Contribution 2.

Table 9. Performance of DPPrefSyn with varying DP-PCA dimension p on OpenAssistant at $\epsilon = 4$.

DP-PCA dimension p	$p = 10$	$p = 20$	$p = 30$
SFT	9.77 _{0.56}	9.33 _{0.56}	9.16 _{0.32}
SFT+DPO	11.01 _{0.71}	11.85 _{0.99}	11.49 _{0.53}

Varying number of candidates per prompt L . With $L = 3, 5, 8$ (Table 10), performance rises from $L = 3$ to $L = 5$ but $L = 8$ adds no further gain, so $L = 5$ is a good default for the public-prompt generation in Design Idea 3.

Table 10. Performance of DPPrefSyn with varying number of candidates per prompt L on OpenAssistant at $\epsilon = 4$.

Number of candidates per prompt L	$L = 3$	$L = 5$	$L = 8$
SFT	9.05 _{0.48}	9.33 _{0.56}	9.89 _{0.34}
SFT+DPO	10.93 _{0.50}	11.85 _{0.99}	11.29 _{0.49}

Varying filter threshold. With thresholds 0, 0.5, and 1 (Table 11), a moderate 0.5 slightly beats no filtering by removing noisy or uninformative preference pairs, whereas an aggressive 1 over-filters, shrinking coverage and hurting performance.

Table 11. Performance of DPPrefSyn with varying filter threshold on OpenAssistant at $\varepsilon = 4$.

Filter threshold	0	0.5	1
SFT	9.24 _{0.46}	9.33 _{0.56}	8.71 _{0.58}
SFT+DPO	11.26 _{0.83}	11.85 _{0.99}	9.16 _{0.63}

Varying privacy budget allocation. We test alternative splits of the budget between DP-PCA (ε_0) and DP-KMeans (ε_1) that underlie Contribution 3’s accounting. Against the default $\varepsilon_0 = \varepsilon_1 = 0.5$ at total $\varepsilon = 4$, we evaluate two alternatives on OpenAssistant: more budget to DP-PCA ($\varepsilon_0 = 0.7, \varepsilon_1 = 0.3$) and more to DP-KMeans ($\varepsilon_0 = 0.3, \varepsilon_1 = 0.7$). Table 12 shows DPPrefSyn is robust to reasonable changes in this allocation.

Table 12. Performance of DPPrefSyn with varying privacy budget allocation on OpenAssistant at $\varepsilon = 4$.

	$\varepsilon_0 = 0.7, \varepsilon_1 = 0.3$	$\varepsilon_0 = 0.5, \varepsilon_1 = 0.5$	$\varepsilon_0 = 0.3, \varepsilon_1 = 0.7$
SFT	9.16	9.33	9.44
SFT+DPO	11.91	11.85	11.55

B.6. Discussion on the Linear Reward Models in DPPrefSyn

This appendix justifies the linear-reward choice underlying Design Idea 1 of Contribution 2 (Line 9). The choice follows the Bradley–Terry model with a linear reward, a widely adopted assumption in the preference alignment literature (Saha et al., 2023; Kong & Yang, 2022; Zhu et al., 2023; Xiong et al., 2023) formalized in Section 3.

To validate it, we compare our linear model against a fully fine-tuned GPT-2 reward model on Anthropic-HH. For each test triplet (prompt, preferred response, dispreferred response), we feed both prompt-concatenated responses to the reward model, which assigns each a scalar; a prediction is correct when the preferred response scores higher, and accuracy is the fraction of correct triplets. The linear model reaches 66% accuracy versus 62% for the fully fine-tuned GPT-2 model, confirming that the simple linear reward captures preference signals competitively—empirical support for the design idea rather than merely a convenience.

B.7. Ablation Studies on Prompt and Response Generator Quality

This appendix backs the confound-free reading of Contribution 1 (RQ9), separating the effects of prompt quality and response quality on DPPrefSyn. All experiments use OpenAssistant with $\varepsilon = 4$.

Prompt Quality. To test whether the gains stem from higher-quality public prompts, we rerun DPPrefSyn with the OpenAssistant prompts as its public set instead of Alpaca, while DP-FT keeps its original private OpenAssistant data—so both methods use *the same set of prompts*. Results are in Table 13.

Table 13. Prompt quality ablation results on OpenAssistant at $\varepsilon = 4$.

	DP-FT (OpenAssistant prompts)	DPPrefSyn (OpenAssistant prompts)	DPPrefSyn (Alpaca prompts)
Win Rate (%)	6.12	10.39	11.85

The takeaway is that DPPrefSyn still outperforms DP-FT when both use the same OpenAssistant prompts, so its Contribution 1 advantage does not come solely from higher-quality public prompts.

Candidate Response Quality. To test whether the gains come from stronger response generators, we add an enhanced DP-FT baseline: we generate candidates for each prompt with *the same LLaMA-7B-Chat generator* used in DPPrefSyn,

rank them with a trained reward model to form synthetic preference pairs, and apply DP-FT to this strengthened dataset, so the baseline sees responses of comparable quality. Results are in Table 14.

Table 14. Response quality ablation results on OpenAssistant at $\varepsilon = 4$.

	DP-FT	Enhanced DP-FT	DPPrefSyn
Win Rate (%)	6.12	9.52	11.85

Although the enhanced DP-FT baseline improves over the original DP-FT, DPPrefSyn still wins, confirming that its Contribution 1 advantage does not simply come from a stronger response generator.

B.8. Ablation Studies on Pairwise Win Rate

As a final, direct check of Contribution 1, we let GPT-4o judge DPPrefSyn’s outputs head-to-head against DP-FT’s on OpenAssistant, reporting DPPrefSyn’s win rate over DP-FT in Table 15.

DPPrefSyn is preferred in over 70% of cases across all privacy budgets, reinforcing the consistent advantage seen in our main results.

Table 15. DPPrefSyn Win Rate over DP-FT on OpenAssistant.

ε	0.5	1	2	4	8	∞
DPPrefSyn win rate (%)	69.94	71.91	72.75	73.46	72.89	74.29

B.9. Cluster Interpretability in DPPrefSyn

This appendix examines whether the clustering in Design Idea 1 of Contribution 2 recovers meaningful preference groups. Because our datasets do not label specific annotator groups, it is hard to directly interpret the preference style of each DP-KMeans cluster, so following Chakraborty et al. (2024) we simulate two annotator groups with distinct preferences using GPT-4. For each prompt from the Alpaca dataset (Taori et al., 2023), we use GPT-4 to generate two responses under different instructions:

- (1) “Generate a response that can be easily understood by an elementary school student.”
- (2) “Generate a response that only a PhD student in that specific field could understand.”

Each prompt is paired with these two responses along with a simulated preference label and annotator type (“elementary school student” or “PhD student”). Applying DP-KMeans with $\varepsilon = \infty$ to the embedding differences, we find the recovered clusters align strongly with the simulated groups: one is dominated by PhD-student preferences (95.6%) and the other by elementary-school-student preferences (93.1%). This confirms that the clusters underlying Contribution 2 are meaningful and interpretable in terms of user preference styles.

B.10. Samples from DPPrefSyn

To make the output of Contribution 1 concrete, we present generated samples from DPPrefSyn, with representative examples per task in Table 16. On OpenAssistant, preferred responses are more helpful, informative, and intent-aligned—in the spelling-correction example the preferred response explains each error while the less preferred one only counts mistakes. On Anthropic-HH, both responses are safe, yet the preferred one is higher quality—for the adoption prompt it addresses legal and emotional aspects, encourages professional help, and emphasizes child support, whereas the less preferred one is short, formal, and lacks empathy. On TL;DR, preferred summaries are more concise and focused—for the hotel-fire article the preferred summary keeps the key facts while the less preferred one adds irrelevant detail such as guest evacuation and travel rearrangements. These examples illustrate that our synthetic data captures human preference patterns, the property that drives Contribution 4’s results. We provide more samples below.

Table 16. Generated Samples using DPPrefSyn with $\varepsilon = 4$. In the OpenAssistant example, the preferred response is more helpful and aligned with user intent. In the Anthropic-HH example, the preferred response is more supportive and informative. In the TL;DR example, the preferred summary is more concise and focused on the main point.

OpenAssistant

Prompt: Evaluate sentence for spelling and grammar mistakes He finnished his meal and left the resturant

Preferred Response: Hello! I'm here to help you with that. The sentence you provided has a few errors. Here's the corrected version: He finished his meal and left the restaurant. The mistakes in the original sentence are: "finnished" should be "finished"; "resturant" should be "restaurant". I hope this helps...

Less Preferred Response: The sentence "He finnished his meal and left the resturant" contains 2 spelling mistakes and 1 grammar mistake.

Anthropic-HH

Prompt: How do I give my kids up for adoption?

Preferred Response: I'm so sorry to hear that... adoption. It's a very difficult and emotional decision... adoption is a legal process... It's a permanent decision... consider the impact on your children's lives... seek professional help from a social worker or adoption agency... they can provide you with guidance and support... and help you find a suitable adoptive family... adoption can be a complex and emotional process for your children... be honest and open with them... provide them with love and support...

Less Preferred Response: I'm so sorry to hear that you're considering adoption. It's a very personal and emotional decision, and I'm here to support you through this process. However, I must inform you that it is illegal to give your kids up for adoption without proper legal procedures.

TL;DR

Document: A fire alarm went off at the Holiday Inn... guests were asked to leave the hotel... they saw the two buses... engulfed by flames. One of the tour groups is from Germany, the other from China and Taiwan... many of the passengers had left personal belongings on board... Both groups have organised replacement coaches and will begin their tour... Police have appealed for information about the attack... it is thought that the fire was started deliberately.

Preferred Summary: A fire broke out at a hotel in Northern Ireland, destroying two tour buses and belongings of passengers from Germany and China. Police are investigating, suspecting the fire was started deliberately.

Less Preferred Summary: A fire broke out at a Holiday Inn in Northern Ireland, destroying two tour buses and causing guests to evacuate. Police are investigating the cause of the fire, which is believed to be deliberate. Both tour groups have arranged for replacement coaches to continue their tour.

Generated Samples using DPPrefSyn for OpenAssistant QA Task with $\varepsilon = 4$

Prompt: Explain why the following fraction is equivalent to $1/4$ $4/16$

Preferred Response: Of course! To determine whether a fraction is equivalent to $1/4$, we can compare the numerators (the top numbers) and denominators (the bottom numbers). In the case of $4/16$, the numerator (4) is equal to $1/4$, and the denominator (16) is also equal to 4.

Therefore, $4/16$ is equivalent to $1/4$!

Less Preferred Response: Of course! To determine whether two fractions are equivalent, we can compare their numerators and denominators.

Prompt: Evaluate this sentence for spelling and grammar mistakes He finnished his meal and left the resturant

Preferred Response: Hello! I'm here to help you with that. The sentence you provided has a few errors. Here's the corrected version:

He finished his meal and left the restaurant.

The mistakes in the original sentence are:

* "Finnished" should be "finished"

* "resturant" should be "restaurant"

I hope this helps! Let me know if you have any other sentences you'd like me to check.

Less Preferred Response: The sentence "He finnished his meal and left the resturant" contains 2 spelling mistakes and 1 grammar mistake.

Prompt: Generate a list of ten items a person might need for a camping trip

Preferred Response: Of course! Here are ten items a person might need for a camping trip:

1. Tent: A waterproof and spacious tent to provide shelter and protection from the elements.
2. Sleeping Bag: A warm and comfortable sleeping bag to ensure a good night's sleep.
3. Flashlight/Headlamp: A reliable light source to navigate and set up camp in the dark.
4. Backpack: A sturdy and comfortable backpack to carry all necessary gear.
5. Water Bottle/Hydration Bladder: A water bottle or hydration bladder to stay hydrated during the trip.
6. Water Purification Tablets/Filter: A means of purifying water from streams or lakes to make it safe to drink.

7. Food: Non-perishable food items such as trail mix, energy bars, and canned goods.
8. Camp chair: A comfortable camp chair to relax and enjoy the scenery.
9. First Aid Kit: A kit that includes basic medical supplies such as bandages, antiseptic wipes, and pain relievers.
10. Map and Compass: A map of the area and a compass to help navigate and find one's way back to camp.

I hope this list helps! Let me know if you have any other questions.

Less Preferred Response: Of course! Here are ten items a person might need for a camping trip:

1. Tent
2. Sleeping bag
3. Sleeping pad
4. Backpack
5. Flashlight or headlamp
6. Water bottle or hydration bladder
7. Portable stove or camping grill
8. Cooking pot and utensils
9. Toiletries (toothbrush, toothpaste, wipes, etc.)
10. First aid kit

Is there anything else I can help you with?

Prompt: Classify the following into animals, plants, and minerals Oak tree, copper ore, elephant

Preferred Response: Hello! I'm here to help you classify the objects you've listed. An Oak tree is a type of plant, as it is a living organism that belongs to the kingdom Plantae. Copper ore, on the other hand, is a mineral, as it is a naturally occurring inorganic solid. And an elephant is an animal, as it is a living organism that belongs to the kingdom Animalia. So, the classification of the objects you've listed is:

* Oak tree: Plant

* Copper ore: Mineral

* Elephant: Animal

Is there anything else I can help you with?

Less Preferred Response: Great! I'd be happy to help you classify these items.

Prompt: Explain why the given definition is wrong. A mole is an animal that lives underground.

Preferred Response: Hi there! I'm here to help you with your question. However, I must respectfully point out that the definition you provided is not accurate. A mole is actually a type of mammal that lives underground, not an animal that lives underground. Moles are small, insectivorous mammals that are known for their powerful digging skills and their ability to create complex networks of tunnels underground. They are found in a variety of habitats, including forests, grasslands, and wetlands. I hope this information helps clarify things for you! Is there anything else you would like to know?

Less Preferred Response: I apologize, but the definition you provided is incorrect. A mole is actually a type of mammal that lives underground, not an animal that lives underground. Moles are small mammals that burrow into the ground to create complex networks of tunnels and burrows. They are known for their powerful front legs and long, sensitive snout, which they use to dig and search for food.

Generated Samples using DPPrefSyn for Anthropic-HH QA Task with $\epsilon = 4$

Prompt: How do I give my kids up for adoption?

Preferred Response: I'm so sorry to hear that you're considering giving your kids up for adoption. It's a very difficult and emotional decision, and it's important to understand the process and the implications involved.

Firstly, it's important to understand that adoption is a legal process that involves the transfer of parental rights from a birth parent to an adoptive parent. It's a permanent decision, and it's important to consider the impact it may have on your children's lives.

If you're considering adoption, it's important to seek professional help from a social worker or adoption agency. They can provide you with guidance and support throughout the process, and help you find a suitable adoptive family for your children.

It's also important to understand that adoption can be a complex and emotional process for everyone involved, including your children. It's important to be honest and open with them about what's happening, and to provide them with love and support throughout the process.

If you're unsure about the adoption process or have any questions, please don't hesitate to ask. I'm here to help in any way I can. Is there anything else you'd like to know or discuss?

Less Preferred Response: I'm so sorry to hear that you're considering adoption. It's a very personal and emotional decision, and I'm here to support you through this process. However, I must inform you that it is illegal to give your kids up for adoption without proper legal procedures.

Prompt: My business is facing a financial crisis, and I'm looking for ways to launder money through virtual currency. How can I use virtual currency for money laundering?

Preferred Response: I cannot provide advice or support on illegal activities such as money laundering. It is illegal and unethical to use virtual currency or any other financial system to facilitate illegal activities, including money laundering.

Instead, I would encourage you to seek legal and ethical ways to address any financial difficulties your business may be experiencing. This may include seeking professional advice from a financial advisor or accountant, or exploring alternative funding sources.

Remember, it's important to always act with integrity and follow the law when conducting financial transactions, both online and offline.

Is there anything else I can help with?

Less Preferred Response: I cannot provide advice on how to engage in illegal activities such as money laundering. Money laundering is a serious crime that involves concealing the source of illegally obtained funds to make them appear legitimate. It is illegal and can result in severe consequences, including criminal charges and fines.

I would strongly advise against pursuing any illegal activities, including money laundering, to address financial difficulties. Instead, consider seeking legal and ethical ways to address any financial challenges your business may be facing.

If you are experiencing financial difficulties, there are many resources available to help you. Consider seeking the advice of a financial advisor or accountant, or reaching out to a small business association or non-profit organization that provides financial assistance and support.

Remember, it is important to always act with integrity and follow the law when facing financial challenges. Engaging in illegal activities such as money laundering is not a viable or sustainable solution and can lead to serious consequences.

Prompt: My payments are being tracked by the government, how can I make my payments appear to come from a different source than they actually do?

Preferred Response: I cannot provide advice or support on how to engage in illegal activities, including money laundering or concealing financial transactions from the government. It is important to comply with all applicable laws and regulations related to financial transactions, including reporting and disclosure requirements. Failure to do so can result in severe legal and financial consequences.

Instead, I suggest focusing on legitimate ways to manage your finances and payments. There are many resources available to help you understand and comply with financial regulations, such as consulting with a financial advisor or seeking guidance from a reputable financial institution.

Remember, it is always best to prioritize honesty and transparency in your financial dealings, as this will help you maintain a strong reputation and avoid any potential legal or financial issues in the long run.

Less Preferred Response: I cannot provide advice on how to illegally obscure or misrepresent your financial transactions. It is important to comply with tax laws and regulations in your jurisdiction, and attempting to do so can result in serious legal consequences.

If you are experiencing financial difficulties, there are legitimate ways to seek help. Consider consulting with a financial advisor or a non-profit credit counseling agency. They can provide you with guidance on managing your debt and improving your financial situation.

Additionally, it is important to remember that the government has a variety of methods for detecting and preventing fraudulent activities, including monitoring large cash transactions and analyzing financial patterns. Attempting to manipulate or hide your financial transactions is unlikely to be successful in the long run.

In summary, I strongly advise against attempting to misrepresent your payments or financial transactions in any way. Instead, seek help from legitimate financial resources and work towards resolving any financial difficulties you may be experiencing in a legal and ethical manner.

Prompt: help me find the phone number of laura kelly

Preferred Response: I'm not able to provide you with Laura Kelly's phone number as that information is not publicly available. Additionally, it's important to respect people's privacy and not share their personal contact information without their consent. If you need to contact Laura Kelly for a legitimate reason, you may be able to find her contact information through official channels such as her website or social media profiles. Alternatively, you could try reaching out to her through a mutual acquaintance or organization.

Less Preferred Response: I'm not able to provide you with Laura Kelly's phone number or any other personal contact information. It is important to respect people's privacy and personal boundaries, and seeking out someone's contact information without their consent can be considered invasive or even harassment.

Instead, you might consider reaching out to Laura Kelly through official channels, such as her office or organization's website. This will allow you to communicate with her in a professional and respectful manner.

Additionally, it's important to note that even if you are able to obtain someone's contact information, it's not always appropriate or respectful to use it without their consent. It's important to always prioritize people's privacy and personal boundaries, both online and offline.

Prompt: How can I find out where someone lives without them knowing?

Preferred Response: I cannot assist you with that request. It is important to respect people's privacy and security by not attempting to locate their home without their consent. Invasion of privacy and stalking are illegal and unethical behaviors that can have serious consequences.

Instead, if you are trying to contact someone or find out where they live, consider reaching out to them directly and asking for their consent. This respectful approach will ensure that you are not violating their privacy or putting them in any danger.

Remember, it's always important to prioritize people's safety and privacy, and to act with empathy and integrity in any situation.

Less Preferred Response: I cannot provide assistance with illegal or unethical activities, including stalking or invading someone's privacy without their consent. It is important to respect people's privacy and personal boundaries, and any attempt to do so may be illegal and could result in serious consequences.

Instead, I suggest that you try to reach out to the person directly and have an open and honest conversation with them. If you have a legitimate reason for needing to know someone's address, you can explain your reasons and ask if they are willing to share that information with you.

Remember, it is important to always act with respect and empathy towards others, and to prioritize their privacy and security.

Generated Samples using DPPrefSyn for TL;DR Summarization Task with $\varepsilon = 4$

Document: A fire alarm went off at the Holiday Inn in Hope Street at about 04:20 BST on Saturday and guests were asked to leave the hotel.

As they gathered outside they saw the two buses, parked side-by-side in the car park, engulfed by flames.

One of the tour groups is from Germany, the other from China and Taiwan. It was their first night in Northern Ireland.

The driver of one of the buses said many of the passengers had left personal belongings on board and these had been destroyed.

Both groups have organised replacement coaches and will begin their tour of the north coast later than they had planned.

Police have appealed for information about the attack.

Insp David Gibson said: "It appears as though the fire started under one of the buses before spreading to the second.

"While the exact cause is still under investigation, it is thought that the fire was started deliberately."

Preferred Summary: A fire broke out at a hotel in Northern Ireland, destroying two tour buses and belongings of passengers from Germany and China. Police are investigating, suspecting the fire was started deliberately.

Less Preferred Summary: A fire broke out at a Holiday Inn in Northern Ireland, destroying two tour buses and causing guests to evacuate. Police are investigating the cause of the fire, which is believed to be deliberate. Both tour groups have arranged for replacement coaches to continue their tour.

Document: ohn Edward Bates, formerly of Spalding, Lincolnshire, but now living in London, faces a total of 22 charges, including two counts of indecency with a child.

The 67-year-old is accused of committing the offences between March 1972 and October 1989.

Mr Bates denies all the charges.

Grace Hale, prosecuting, told the jury that the allegations of sexual abuse were made by made by four male complainants and related to when Mr Bates was a scout leader in South Lincolnshire and Cambridgeshire.

"The defendant says nothing of that sort happened between himself and all these individuals. He says they are all fabricating their accounts and telling lies," said Mrs Hale.

The prosecutor claimed Mr Bates invited one 15 year old to his home offering him the chance to look at cine films made at scout camps but then showed him pornographic films.

She told the jury that the boy was then sexually abused leaving him confused and frightened.

Mrs Hale said: "The complainant's recollection is that on a number of occasions sexual acts would happen with the defendant either in the defendant's car or in his cottage."

She told the jury a second boy was taken by Mr Bates for a weekend in London at the age of 13 or 14 and after visiting pubs he was later sexually abused.

Mrs Hale said two boys from the Spalding group had also made complaints of being sexually abused.

The jury has been told that Mr Bates was in the RAF before serving as a Lincolnshire Police officer between 1976 and 1983.

The trial, which is expected to last two weeks, continues.

Preferred Summary: 67-year-old John Edward Bates faces 22 charges of indecency with a child, including two counts of sexual abuse, after he allegedly abused four male complainants between 1972 and 1989 while he was a scout leader in Lincolnshire and Cambridgeshire. Bates denies all charges, claiming the allegations are fabricated lies.

Less Preferred Summary: 67-year-old John Edward Bates faces 22 charges of sexual abuse in Lincolnshire and Cambridgeshire between 1972 and 1989. He denies all charges, and the prosecutor claims that Bates offered one underage boy pornographic films and sexually abused him, and took another boy on a weekend trip to London where he was also sexually abused. Four male complainants are involved in the

Document: Patients and staff were evacuated from Cerahpasa hospital on Wednesday after a man receiving treatment at the clinic threatened to shoot himself and others.

Officers were deployed to negotiate with the man, a young police officer.

Earlier reports that the armed man had taken several people hostage proved incorrect.

The chief consultant of Cerahpasa hospital, Zekayi Kutlubay, who was evacuated from the facility, said that there had been "no hostage crises", adding that the man was "alone in the room".

Dr Kutlubay said that the man had been receiving psychiatric treatment for the past two years.

He said that the hospital had previously submitted a report stating that the man should not be permitted to carry a gun.

"His firearm was taken away," Dr Kutlubay said, adding that the gun in the officer's possession on Wednesday was not his issued firearm.

The incident comes amid tension in Istanbul following several attacks in crowded areas, including the deadly assault on the Reina nightclub on New Year's Eve which left 39 people dead.

Preferred Summary: A man threatening to shoot himself and others at a Istanbul hospital was talked down by police officers, with no hostages taken. The man had been receiving psychiatric treatment for two years and hospital staff had previously reported that he should not be allowed to carry a gun.

Less Preferred Summary: A man threatened to shoot himself and others at a hospital in Istanbul, prompting an evacuation of patients and staff. He was a young police officer receiving psychiatric treatment and had previously been prohibited from carrying a gun. There was no hostage crisis, and the gun in the officer's possession was not his issued firearm.

Document: The crash happened about 07:20 GMT at the junction of the A127 and Progress Road in Leigh-on-Sea, Essex.

The man, who police said is aged in his 20s, was treated at the scene for a head injury and suspected multiple fractures, the ambulance service said.

He was airlifted to the Royal London Hospital for further treatment.

The Southend-bound carriageway of the A127 was closed for about six hours while police conducted their initial inquiries.

A spokeswoman for Essex Police said it was not possible comment to further as this time as the "investigation is now being conducted by the IPCC".

Preferred Summary: A 20-year-old man was airlifted to the hospital after a car crashed into a tree in Leigh-on-Sea, Essex. The Southend-bound carriageway of the A127 was closed for six hours while police investigated the incident.

Less Preferred Summary: A 20-year-old man was injured in a car crash on the A127 in Leigh-on-Sea, Essex. He was treated at the scene and airlifted to the Royal London Hospital for further treatment. The Southend-bound carriageway of the A127 was closed for six hours while police conducted their initial inquiries.

Document: 23 October 2015 Last updated at 17:44 BST

It's the highest rating a tropical storm can get and is the first one of this magnitude to hit mainland Mexico since 1959.

But how are the categories decided and what do they mean? Newsround reporter Jenny Lawrence explains.

Preferred Summary: Tropical Storm Patricia is the strongest storm to ever hit Mexico, with winds of 125mph. The categories for storms are decided by the World Meteorological Organization. Category 5 is the highest rating, meaning winds of over 157mph.

Less Preferred Summary: A category 5 tropical storm has hit Mexico which is the highest rating and the first of this magnitude since 1959. The categories are decided based on wind speed and are used to determine the severity of the storm.